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Module 1.2 Market Failure and Government Intervention

1. What is Market Failure?

Market failure occurs when the free market mechanism leads to a misallocation of resources, resulting in a loss of economic welfare. The market either over-produces or under-produces goods relative to the socially optimal level.

In an efficient market, the price mechanism allocates resources to maximise total social welfare. Market failure means this does not happen. The socially optimal output is where **Marginal Social Benefit (MSB) = Marginal Social Cost (MSC)**.

1.1 Types of Market Failure

- **Externalities** — costs or benefits affecting third parties not involved in a transaction
- **Public goods** — goods the market will not provide (or will under-provide)
- **Merit and demerit goods** — goods under-consumed or over-consumed due to information failure
- **Information failure** — asymmetric or imperfect information distorts decision-making
- **Monopoly power** — dominance leads to higher prices and lower output
- **Factor immobility** — resources cannot move freely between uses
- **Inequality** — the market may produce outcomes deemed inequitable

1.2 Private, External, and Social Costs/Benefits

$$\text{Social Cost} = \text{Private Cost} + \text{External Cost}$$

$$\text{Social Benefit} = \text{Private Benefit} + \text{External Benefit}$$

Private costs/benefits are borne by the economic agents making the decision. **External costs/benefits** fall on third parties. The **social** values combine both.

2. Externalities

Externalities are spillover effects of production or consumption that affect third parties who are not directly involved in the economic transaction. They cause a divergence between private and social costs or benefits.

2.1 Negative Externality of Production

The production process imposes costs on third parties. The social cost exceeds the private cost (MSC > MPC). Producers do not bear the full cost, so they **over-produce** relative to the social optimum.

Examples: Factory pollution contaminating rivers, carbon emissions from power stations, noise from airports, chemical run-off from agriculture.

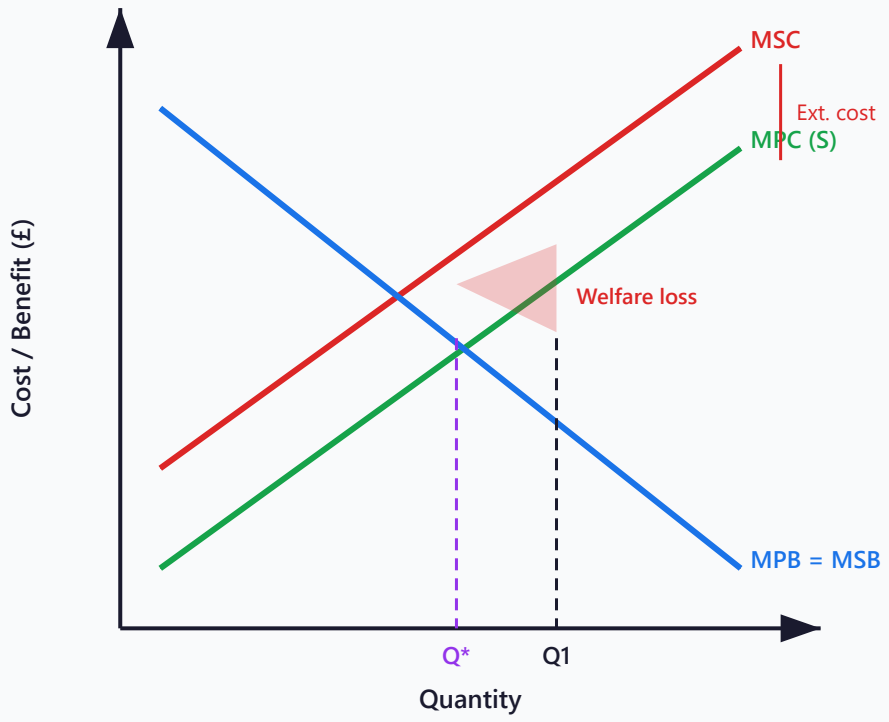

*Figure 1: Negative externality of production — market overproduces at Q1 vs socially optimal Q**

Case Study: Air Pollution from UK Power Stations

Coal-fired power stations in the UK historically imposed significant external costs through sulphur dioxide and CO2 emissions, causing acid rain and contributing to climate change. The social cost of electricity generation far exceeded the private cost. The UK's shift away from coal (from 40% of electricity in 2012 to 0% by 2024) was driven by carbon pricing through the EU ETS and the UK Carbon Price Support, which internalised external costs, making coal uneconomic relative to gas and renewables.

2.2 Negative Externality of Consumption

Consumption imposes costs on third parties. The social benefit is less than the private benefit (MSB < MPB). Consumers **over-consume** because they do not account for the costs they impose on others.

Examples: Smoking (passive smoking, NHS costs), alcohol (antisocial behaviour, healthcare), driving cars (pollution, congestion).

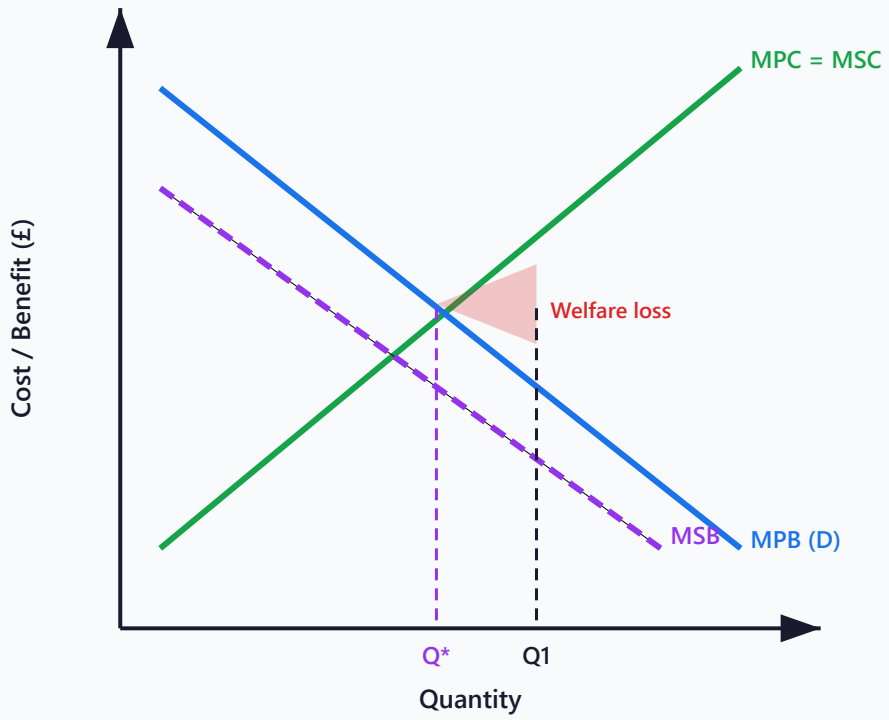

Figure 2: Negative externality of consumption — MSB below MPB, overconsumption at Q1

2.3 Positive Externality of Production

Production creates benefits for third parties. The social cost is less than the private cost (MSC < MPC). The market **under-produces** because producers do not capture the full social benefit.

Examples: A beekeeper's bees pollinating neighbouring farms, R&D spillovers, training workers who then move to other firms.

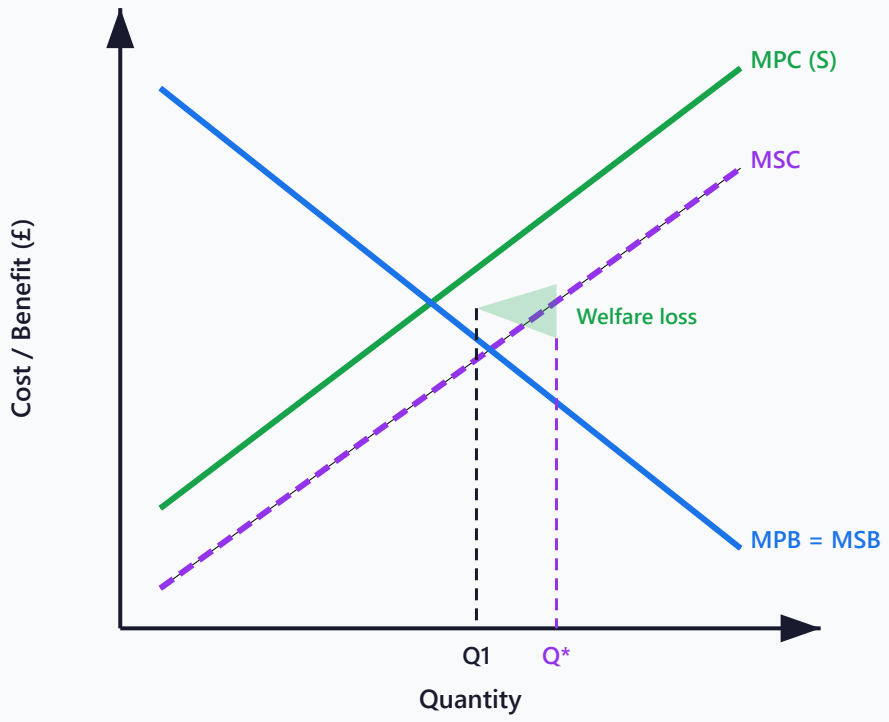

*Figure 3: Positive externality of production — underproduction at Q1 vs social optimum Q**

2.4 Positive Externality of Consumption

Consumption creates benefits for third parties. The social benefit exceeds the private benefit (MSB > MPB). The market **under-consumes**.

Examples: Education (higher productivity benefits employers and the economy), vaccination (herd immunity), public transport use (reduces congestion for others).

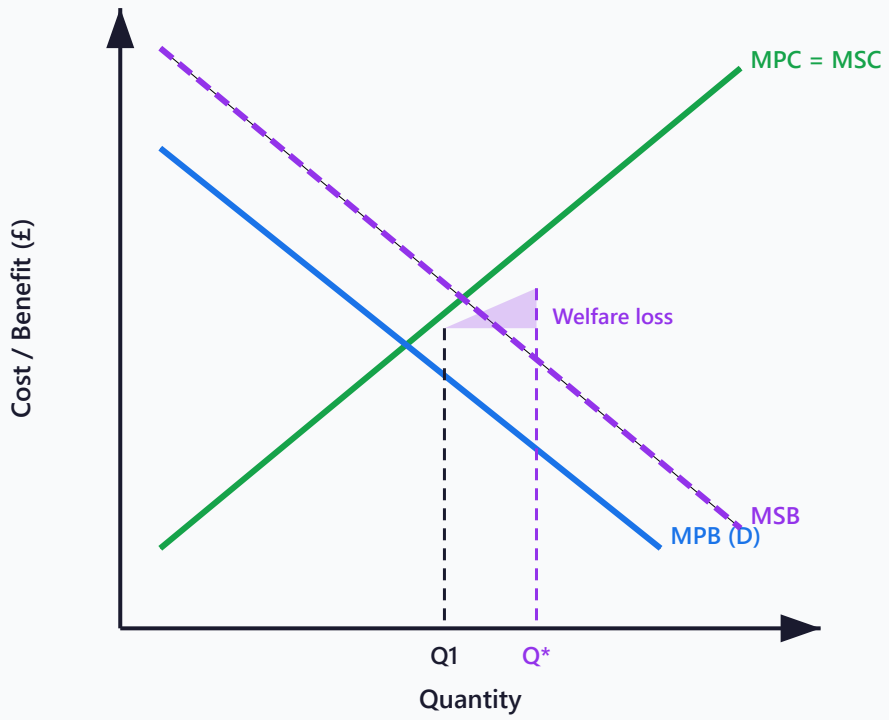

Figure 4: Positive externality of consumption — MSB above MPB, underconsumption at Q1

Case Study: COVID-19 Vaccination Programme

COVID-19 vaccination is a textbook case of positive consumption externalities. When an individual gets vaccinated, they protect themselves (private benefit) and reduce transmission to others (external benefit), contributing to herd immunity. The UK government provided vaccines free at the point of use, effectively subsidising consumption to the social optimum. By early 2022, over 90% of UK adults had received at least one dose. Without intervention, vaccination rates would have been substantially lower, and the free market would have underproduced this good with major positive externalities.

Exam tip: In externality diagrams, always label MPC, MSC, MPB, MSB, Q1 (market output), Q* (socially optimal output), and shade the welfare loss triangle. State clearly whether the good is over- or under-produced/consumed.

3. Public Goods

Public goods are goods that are **non-excludable** (impossible to prevent non-payers from consuming) and **non-rivalrous** (one person's consumption does not reduce availability for others). The free market will fail to provide them — a **missing market**.

3.1 Key Characteristics

Characteristic	Meaning	Implication
Non-excludable	Cannot prevent non-payers from consuming	Free-rider problem — rational individuals will not pay voluntarily
Non-rivalrous	One person's consumption does not diminish availability	Marginal cost of extra user is zero — inefficient to ration by price

3.2 The Free-Rider Problem

Because public goods are non-excludable, individuals can consume without paying — they "free ride." Since everyone has this incentive, no one voluntarily pays, so private firms cannot charge a price and will not supply the good. This is a **complete market failure**.

Solution: The government must provide public goods, funded through compulsory taxation.

3.3 Examples of Public and Quasi-Public Goods

Good	Non-excludable?	Non-rivalrous?	Classification
National defence	Yes	Yes	Pure public good
Street lighting	Yes	Yes	Pure public good
Flood defences	Yes	Yes	Pure public good
Roads	Mostly yes (but toll roads)	Rivalrous when congested	Quasi-public good

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Parks	Mostly yes (some charge)	Rivalrous when crowded	Quasi-public good
BBC broadcasts	Excludable (licence)	Non-rivalrous	Quasi-public good

4. Merit and Demerit Goods

4.1 Merit Goods

> **Merit goods** are goods that are under-consumed in a free market because consumers underestimate the private benefits and/or because of positive externalities.

Under-consumed for two reasons:

1. **Information failure:** Consumers do not fully appreciate the benefits (e.g. undervaluing education, preventive healthcare).
2. **Positive externalities:** Consumption generates benefits for third parties that consumers do not consider.

Examples: Education, healthcare, museums, exercise, vaccinations, healthy food.

Government response: Subsidise provision, provide free at point of use (NHS, state schools), legislate (compulsory education to age 18), provide information.

4.2 Demerit Goods

> **Demerit goods** are goods that are over-consumed in a free market because consumers overestimate private benefits (or underestimate private costs) and/or because of negative externalities.

Over-consumed for two reasons:

1. **Information failure:** Consumers do not fully understand harm (e.g. young smokers underestimate addiction risk).

2. **Negative externalities:** Consumption imposes costs on third parties (passive smoking, alcohol-fuelled crime).

Examples: Tobacco, alcohol, recreational drugs, gambling, fast food.

Government response: Tax (tobacco duty, alcohol duty), regulate (age restrictions, advertising bans), ban (illegal drugs), provide information (health warnings).

> **Exam tip:** Merit/demerit goods involve a **value judgement** — who decides what is "good" or "bad"? This is inherently paternalistic. Always include this as an evaluation point.

5. Information Failure

> **Information failure** occurs when one party in a transaction has more or better information than the other (asymmetric information), or when consumers/producers lack information needed to make rational decisions (imperfect information).

5.1 Types of Information Failure

- **Asymmetric information:** One party knows more than the other (e.g. a seller knows a car's hidden faults; a patient knows less than their doctor).
- **Imperfect information:** Neither party has full knowledge (e.g. long-term health effects of a new food additive).

5.2 Consequences

- **Adverse selection:** In insurance, high-risk individuals are most likely to buy cover, driving up premiums and potentially causing market collapse (Akerlof's "Market for Lemons").
- **Moral hazard:** Once insured, individuals may take greater risks because they do not bear the full cost.
- **Under-consumption of merit goods** and **over-consumption of demerit goods** due to poor information.
- **Misallocation of resources:** Resources flow to products consumers overvalue and away from those they undervalue.

Case Study: Akerlof's Market for Lemons (1970)

George Akerlof demonstrated how asymmetric information can destroy markets. In used car markets, sellers know if their car is good or a "lemon" (poor quality), but buyers cannot tell. Buyers offer a price reflecting average quality. Good-car owners withdraw (unable to get a fair price), leaving only lemons. This adverse selection spirals until the market collapses. Real-world solutions include warranties, vehicle history checks (HPI in the UK), dealer reputation systems, and independent inspections — all reducing the information gap.

6. Government Intervention to Correct Market Failure

6.1 Indirect Taxes

Indirect taxes are taxes on goods/services that raise costs of production, shifting supply left. They aim to internalise negative externalities by making private cost equal social cost.

- **Specific tax:** Fixed amount per unit (e.g. 58p per litre on petrol). Parallel upward shift of supply curve.
- **Ad valorem tax:** Percentage of price (e.g. VAT at 20%). Supply curve pivots upward (larger absolute shift at higher prices).

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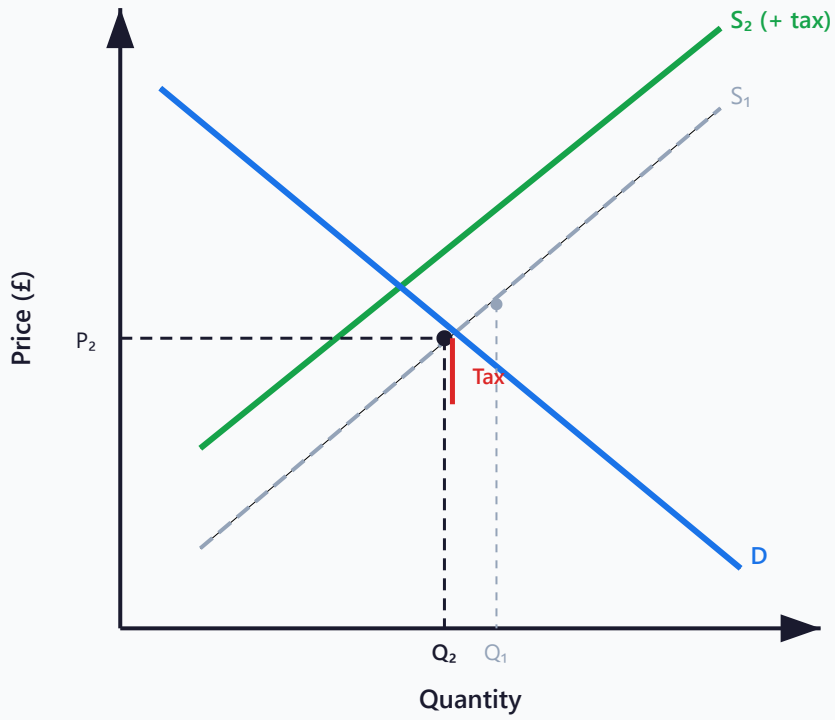

Figure 5: Effect of a specific indirect tax — supply shifts left, price rises, quantity falls

Tax Incidence

The burden of a tax is shared between consumers (who pay a higher price) and producers (who receive less after tax). The **incidence** depends on elasticity:

- If demand is **inelastic**, consumers bear most of the tax burden (price rises significantly, quantity barely falls).
- If demand is **elastic**, producers bear more of the burden (they cannot raise prices much without losing sales).

Evaluation of Indirect Taxes:

- + Internalise externalities, generate revenue, create price signal to change behaviour
- - Difficult to calculate exact external cost, may be regressive, can encourage black markets, demand may be too inelastic for significant effect, firms may relocate

6.2 Subsidies

Subsidies are government payments to producers that reduce production costs, shifting supply right. They aim to increase consumption of goods with positive externalities or merit goods.

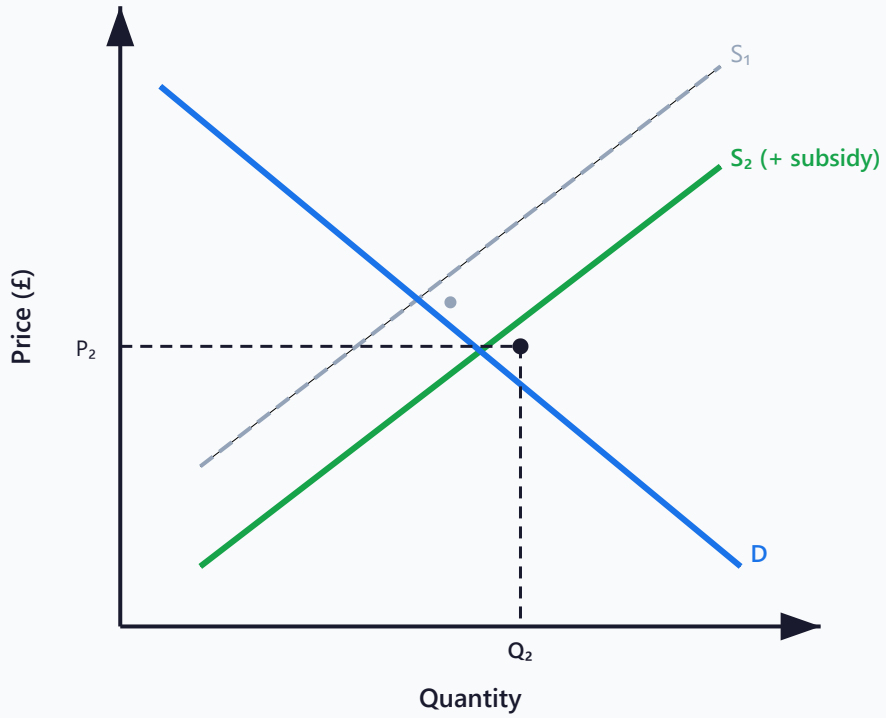

Figure 6: Effect of a subsidy — supply shifts right, price falls, quantity rises

Evaluation of Subsidies:

- + Increase consumption/production of beneficial goods, lower prices, target specific markets
- - Opportunity cost of spending, firms become subsidy-dependent, difficult to set correct level, may benefit producers more if supply inelastic, can distort markets

6.3 Maximum and Minimum Prices

Maximum Price (Price Ceiling)

Set **below equilibrium** to make goods affordable. Example: rent controls.

Effects: Excess demand (shortage), possible black markets, rationing needed, reduced quality, reduced supply (producers exit).

Minimum Price (Price Floor)

Set **above equilibrium** to guarantee producers higher income. Examples: National Minimum Wage, agricultural price floors.

Effects: Excess supply (surplus), government may need to buy surplus, may cause unemployment in labour markets, higher consumer prices.

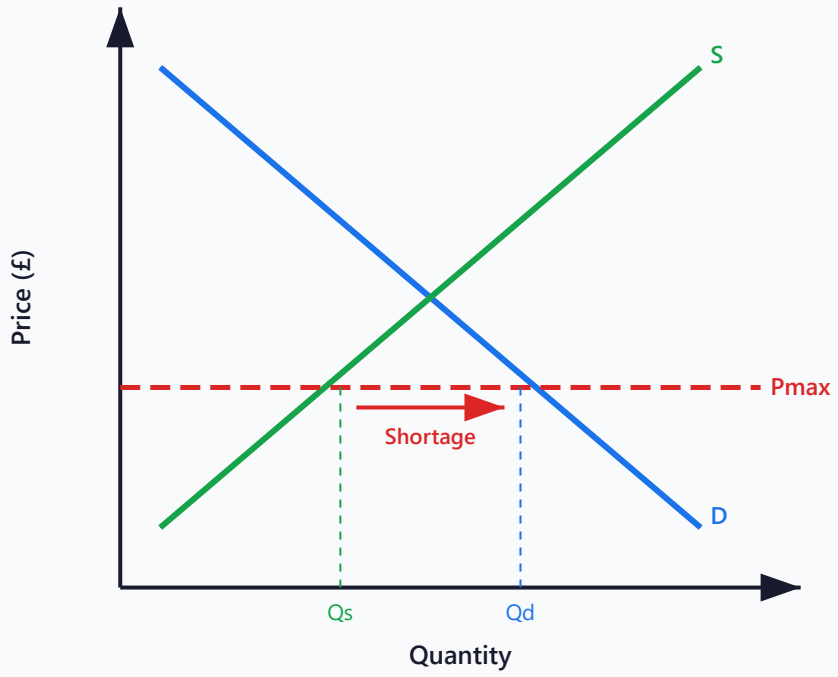

Figure 7: Maximum price (price ceiling) set below equilibrium creates a shortage

6.4 Tradeable Pollution Permits (Cap and Trade)

Tradeable pollution permits: The government sets an overall cap on emissions, issues permits to firms, and allows trading. Firms that reduce emissions below their allocation sell spare permits to those exceeding theirs.

- Creates a financial incentive to cut emissions
- Overall pollution is capped and can be reduced over time
- Most efficient firms benefit; heavy polluters bear higher costs
- Market-based — allows flexibility for firms to choose how to reduce emissions

Case Study: EU Emissions Trading System (EU ETS)

The EU ETS, launched in 2005, is the world's largest carbon market. It covers approximately 40% of EU greenhouse gas emissions from power stations, factories, and airlines. The system initially suffered from government failure — too many free permits were issued, causing the carbon price to collapse to below 5 euros per tonne, providing minimal incentive to reduce emissions. After reforms including the Market Stability Reserve (2019) reducing surplus permits, prices rose above 80 euros per tonne by 2023, finally creating meaningful incentives for decarbonisation. This illustrates both the potential and pitfalls of market-based environmental policy.

6.5 Regulation and Legislation

- **Bans:** Smoking in public places (2007 UK), CFC ban, single-use plastics ban
- **Standards:** Vehicle emissions standards (Euro 6), food safety regulations, building codes
- **Restrictions:** Age limits on alcohol/tobacco, planning permission requirements

Evaluation of Regulation:

- + Can eliminate harmful activities entirely, clear rules, protects vulnerable groups
- - Costly to enforce, may create black markets, inflexible one-size-fits-all, may stifle innovation, risk of regulatory capture

6.6 Provision of Information

Addressing information failure through: health warnings on cigarettes, nutritional labels, energy ratings on appliances, Change4Life campaigns, financial product disclosure requirements.

6.7 State Provision

Government directly provides public goods (defence, street lighting) and merit goods (NHS, state education), funded through taxation, ensuring consumption is not limited by ability to pay.

7. Government Failure

Government failure occurs when government intervention to correct market failure leads to a net welfare loss — making the situation worse or creating new inefficiencies.

7.1 Causes of Government Failure

- **Distortion of price signals:** Subsidies and price controls distort market signals (e.g. agricultural subsidies causing overproduction).
- **Unintended consequences:** Rent controls reducing housing supply; minimum wage potentially causing unemployment.
- **Administrative costs:** Enforcement and bureaucracy may exceed the benefits of intervention.
- **Information gaps:** Government cannot accurately measure externalities or set optimal tax/subsidy levels.
- **Regulatory capture:** Industries influence their regulators to serve producer interests rather than society's.
- **Political short-termism:** Election cycles incentivise vote-winning over welfare-maximising policies.
- **Moral hazard:** Bailouts and safety nets may encourage excessive risk-taking.

7.2 Examples

Intervention	Intended Outcome	Government Failure
Rent controls	Affordable housing	Reduced supply, lower quality, black markets
EU CAP price supports	Food security, farmer income	Overproduction, environmental damage, trade distortion
2008 bank bailouts	Financial stability	Moral hazard — banks took excessive risks expecting rescue

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Biofuel subsidies	Reduce carbon emissions	Diverted agricultural land from food, raising food prices globally

Evaluation — Market Failure vs Government Failure

- Market failure does not automatically justify intervention — the cure may be worse than the disease.
- The key question: does the welfare loss from market failure exceed the welfare loss from government failure?
- The choice is rarely between perfect markets and perfect government — both are imperfect.
- The appropriate intervention depends on the specific market, externality size, and government capacity.
- A combination of interventions (e.g. tax + information + regulation) may be more effective than any single approach.

Exam tip: For high-mark essay questions, always evaluate both the market failure AND the potential for government failure. The best answers consider both sides and reach a supported judgement about whether intervention improves outcomes on balance.

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